

Quiz and midsem makeup

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Course: Stochastic Processes

Q1 a. classify the process in term of State space and ~~the~~ time domain

State space: discrete

Time domain: discrete

b. if claim amounts changes to continuous, then state space will change ~~to~~ to continuous and the time domain remain discrete

Q2. Define (i) Simple symmetric ~~random~~ random walk on the integers
It is defined by $X_n = X_{n-1} + I_n$ where $I_n = +1$ or -1 with equal probability

(ii) Transition probabilities: are the probabilities that the process moves from state i to state j ~~in an~~ in one time step:

$$P(i, i+1) = \frac{1}{2}$$

$$P(i, i-1) = \frac{1}{2}$$

Q3 (a) The process is a Markov. Because the transition probability ~~$P_t = e^{-(t-1)}$~~ $P_t = e^{-(t-1)}$ depends only on the current state X_{t-1}

b. ~~No~~ NO there is no independent increment because the distribution depends on the current state

Q4 A Gamble ruin

$$x_0 = 0$$

$$P_{n+1} = P$$

$$P_{n-1} = 1-P$$

~~x_1, x_0, x_0~~

(1, 2, 1, 2, 3, 4, 5, 6, 5, 6)

Steps

+1, -1, +1, +1, +1, +1, +1, -1, +1

up (+) = 7

down (-) = 2

$\therefore \hat{P} = \frac{7}{9}$ based on the proportion of upward moves

Q5 Given

$$a = \$70$$

$$q = \frac{1}{4}$$

$$p = \frac{3}{4}$$

$$m = 100$$

$$a. f_a = \frac{p^a - p^m}{1 - p^m}$$

$$P = \frac{q}{p} = \frac{1/4}{3/4} = \frac{1}{3}$$

$$p = \frac{1}{3}$$

$$f_a = \frac{\frac{1}{3}^{70} - \frac{1}{3}^{100}}{1 - \frac{1}{3}^{100}}$$

0.0...

$$f_{70} \approx \left(\frac{1}{3}\right)^{70}$$

$$b. da = \frac{1}{q-p} \left(a - m \frac{1-p^q}{1-p^m} \right)$$

$$d_{70} = \frac{1}{\frac{1}{4} - \frac{3}{4}} \left(70 - 100 \cdot \frac{1 - \frac{1}{3}^{70}}{1 - \frac{1}{3}^{100}} \right)$$

$$\begin{aligned} & 70 - 100 \\ & -2(-30) \\ & 60 \end{aligned}$$

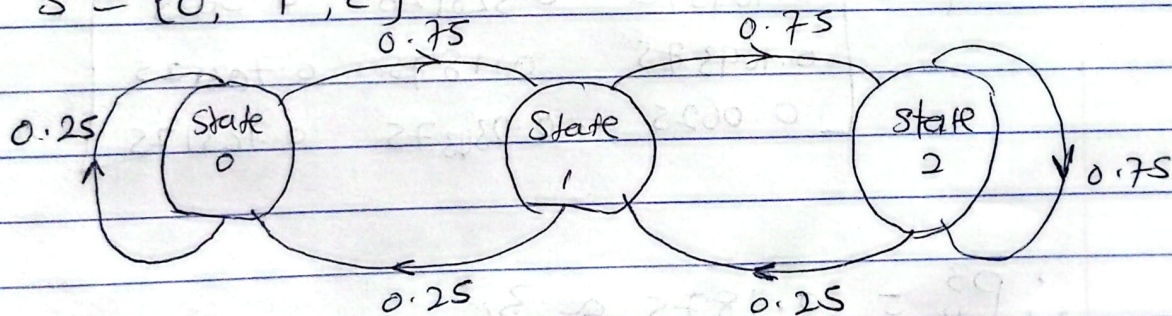
Q6. State space $\{0, 1, 2\}$

$$q_{0,1} = 3 \quad q_{1,0} = 2, \quad q_{1,2} = 4 \quad q_{2,1} = 1$$

(a) generator matrix Q for the process

$$Q = \begin{bmatrix} -3 & 3 & 0 \\ 2 & -6 & 4 \\ 0 & 1 & -1 \end{bmatrix}$$

Q7. $S = \{0, 1, 2\}$



Calculate 3 step probability

$$P = \begin{bmatrix} 0.25 & 0.75 & 0 \\ 0.25 & 0 & 0.75 \\ 0 & 0.25 & 0.75 \end{bmatrix}$$

$$P_{11}^3 = P^3, P^2 = \begin{bmatrix} 0.25 & 0.75 & 0 \\ 0.25 & 0 & 0.75 \\ 0 & 0.25 & 0.75 \end{bmatrix} \begin{bmatrix} 0.25 & 0.75 & 0 \\ 0.25 & 0 & 0.75 \\ 0 & 0.25 & 0.75 \end{bmatrix}$$

$$P^2 = (0.25 \times 0.25) + (0.75 \times 0.25) + (0 \times 0) = 0.25$$

$$(0.25 \times 0.75) + (0.75 \times 0) + (0 \times 0.25) = 0.1875$$

$$(0 \times 0.25) + (0.75 \times 0.75) + (0 \times 0.75) = 0.5625$$

$$P^2 = \begin{bmatrix} 0.25 & 0.1875 & 0.5625 \\ 0.0625 & 0.1875 & 0.5625 \\ 0.0625 & 0.1875 & 0.75 \end{bmatrix}$$

$$P^3 = P \times P^2$$

$$\begin{bmatrix} 0.25 & 0.75 & 0 \\ 0.25 & 0 & 0.75 \\ 0 & 0.25 & 0.75 \end{bmatrix} \begin{bmatrix} 0.25 & 0.1875 & 0.5625 \\ 0.0625 & 0.1875 & 0.5625 \\ 0.0625 & 0.1875 & 0.75 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.109375 & 0.328125 & 0.5625 \\ 0.109375 & 0.1875 & 0.703125 \\ 0.0625 & 0.234375 & 0.703125 \end{bmatrix}$$

$$\therefore P_{11}^3 = 0.1875 \approx \frac{3}{16}$$

Q8 Let $X_n, n \geq 0$
 $S = \{1, 2, 3\}$

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.4 & 0.1 & 0.5 \end{bmatrix}$$

Compute $P_{13}^{(2)}$

$$P_{ij}^{n+m} = \sum_{k=1}^3 P_{ik}^{(n)} P_{kj}^{(m)}$$

$$P_{13}^{(2)} = \sum_{k=1}^3 P_{1k}^{(1)} P_{k3}^{(1)}$$

~~$$P_{13}^{(2)} = P_{11} + P_{13} + P_{12} P_{23}$$~~

$$P_{13}^{(2)} = P_{11} P_{13} + P_{12} P_{23} + P_{13} P_{33}$$

$$0.6(0.1) + 0.3(0.3) + 0.1(0.5)$$

$$P_{13}^{(2)} = 0.2$$